

Bharat Badgujar

Linux module 2

Assignment 8

1.

Install the nfs-kernel-server package on your Linux VM and create a shared directory named /srv/music_share for sharing music files.

Ans : **Install NFS Server**

```
sudo apt update
```

```
sudo apt install nfs-kernel-server -y
```

```
sudo mkdir -p /srv/music_share
```

2.

Configure /etc/exports to share /srv/music_share with read and write permissions to a specific client IP (use your own second VM or localhost for testing), then restart the NFS service.

Ans : **Configure /etc/exports**

Edit:

```
sudo nano /etc/exports
```

Add:

```
/srv/music_share 192.168.1.10(rw,sync,no_subtree_check)
```

(Replace 192.168.1.10 with your client IP or 127.0.0.1 for testing.)

Restart NFS:

```
sudo exportfs -ra
```

```
sudo systemctl restart nfs-kernel-server
```

3.

On a client machine, mount the NFS share from your server to a local directory named ~/mnt/playlist and verify you can create and access files in the shared folder.

Ans : **Mount on Client**

```
mkdir -p ~/mnt/playlist
```

```
sudo mount SERVER_IP:/srv/music_share ~/mnt/playlist
```

Test:

```
touch ~/mnt/playlist/test.txt
```

```
ls ~/mnt/playlist
```

4.

Simulate a Zomato-style shared menu folder: add a subfolder called 'restaurant_menus' inside your NFS share, copy any 2 PDF or image files into it, and confirm they are visible from the client mount.

Ans : **Create restaurant_menus**

```
mkdir /srv/music_share/restaurant_menus
```

```
cp file1.pdf file2.jpg /srv/music_share/restaurant_menus/
```

On the client:

```
ls ~/mnt/playlist/restaurant_menus
```

You should see both files.

5.

Un-mount the NFS share from the client and update your /etc/exports file to restrict access to read-only, then remount and test that you cannot create or modify files in the shared folder.

Hint: Use the 'ro' option in /etc/exports for read-only access.

Ans : **Change to Read-Only**

Unmount:

```
sudo umount ~/mnt/playlist
```

Edit /etc/exports:

```
/srv/music_share 192.168.1.10(ro,sync,no_subtree_check)
```

Apply changes:

```
sudo exportfs -ra
```

```
sudo systemctl restart nfs-kernel-server
```

Remount:

```
sudo mount SERVER_IP:/srv/music_share ~/mnt/playlist
```

Test:

```
touch ~/mnt/playlist/newfile.txt
```

Expected output:

```
touch: cannot touch 'newfile.txt': Read-only file system
```